

Kelowna, Canada, May 13, 2026

Dear Editors,

We are pleased to submit our manuscript entitled “*Rising temperatures alter how and where boreal mammals move*” for consideration as a research article in Global Change Biology. Below are the answers to the questions requested for submission.

1. What scientific question is addressed? How does ambient temperature affect the movement rates and habitat selection of large Nearctic mammals, and how will rising temperatures under climate change alter these behaviours through to 2100?
2. What are the key findings? Temperature drives species-specific, nonlinear changes in movement and habitat selection across six large mammal species. Projections to 2100 show how some species will move more, others less. These changes have consequences for community dynamics, including predator-prey encounter rates, illustrated by a four-fold increase in caribou-wolf co-occupancy between -20°C and 20°C.
3. Why is this work important and timely? Northern regions are experiencing accelerating warming and extreme heat events. Understanding how wildlife will respond to this change is critical, but most prior work focuses on single species in isolation. This multi-species approach provides novel insight into how differing thermal responses will reshape communities and alter predator-prey dynamics.
4. How does it fit GCB’s scope? GCB publishes studies on biological responses and feedbacks to change across levels of organization. This manuscript addresses organismal-level responses to temperature and scales these responses up to community-level dynamics across multiple trophic levels. This directly fitting GCB’s mandate to promote understanding of biological systems and all aspects of environmental change.
5. What biological and global change aspects does it address? This manuscript examines how rising temperatures and shifting phenology alter movement behaviour and habitat selection across a community of large mammals, with cascading consequences for predator-prey dynamics and community structure. Biological responses are projected through 2100 under four IPCC Shared Socioeconomic Pathways.
6. Three most relevant recent papers? Dickie et al. (2024, GCB) identified climate as underpinning ungulate northern range expansion into caribou habitat, with community-level implications. Lessard et al. (2025, Biodiversity and Conservation) demonstrated weather and disturbance interact to shape caribou movement. Liao et al. (2026, GCB) showed how dynamic conditions under climate change alter caribou movement.
7. Provide the name and institutional email addresses of five potential reviewers. Briana Abrahms (abrahms@uw.edu), Laura Prugh (lprugh@uw.edu), Marketa Zimova (mzimova@uwo.edu), John DeLong (jpdelong@unl.edu), and David Boukal (dboukal@prf.jcu.cz)

We thank you for considering our submission and look forward to your response.

Kind regards,

Stefano Mezzini and Prof. Michael Noonan on behalf of all co-authors